

On the pseudospectral implementation of a class of effective collision operators

In this short note, a simple algorithm based on a pseudo-spectral method is put forward for the implementation of the class of collision operators proposed by Morrison [Physica D, 1986].

1 Construction of the operators. Such operators act on a scalar field

$$u : \Omega \rightarrow \mathbb{R},$$

where the domain $\Omega = [0, 2\pi]^2$ is equipped with doubly periodic boundary conditions so that $\Omega \cong \mathbb{T}^2$. The considered operators are obtained from the symmetric brackets [Morrison, Physica D, 1986]

$$(A, B) = - \int_{\Omega} \int_{\Omega} (\nabla A_u(x) - \nabla A_u(x')) \cdot T(x, x') (\nabla B_u(x) - \nabla B_u(x')) dx' dx,$$

for any two functional $u \mapsto A(u) \in \mathbb{R}$ and $u \mapsto B(u) \in \mathbb{R}$, and A_u, B_u are the functional derivatives of A and B with respect to the L^2 -product evaluated at u , respectively, e. g.,

$$DA(u)v = \int_{\Omega} A_u(x)v(x)dx,$$

where DA is the Frechét derivative of A .

The kernel $T(x, x')$ is matrix valued and in general can be of the form,

$$T(x, x') = \frac{1}{2} M(u(x)) M(u(x')) w(x, x'),$$

where the function M and the new kernel $w(x, x')$ have to be chosen properly.

The function M is essentially determined by the chosen entropy functional, as it will be briefly discussed below.

As for the kernel w , we want to construct it so that a given energy functional $H(u)$ satisfies a Casimir-like condition for the symmetric brackets, namely,

$$(A, H) = 0, \quad \text{for all functionals } A.$$

This is achieved by considering the vector-valued function

$$g(x, x') = \nabla H_u(x) - \nabla H_u(x'),$$

and constructing the projector on the orthogonal space of $g(x, x')$ point-wise for all pair (x, x') , namely,

$$w(x, x') = f(x, x') \pi(x, x'),$$

where f can be an arbitrary function of (x, x') , symmetric under exchange $x \leftrightarrow x'$, and

$$\pi(x, x') = I - \frac{g(x, x') \otimes g(x, x')}{|g(x, x')|^2},$$

is the projector, with I the identity matrix. The choice $f = 1/|g|$ is typical for operators such as the Landau collision operator.

As a consequence of this construction, we have the following properties automatically:

1. Symmetry as a matrix,

$$w_{ij}(x, x') = w_{ji}(x, x');$$

2. Symmetry as a kernel,

$$w_{ij}(x, x') = w_{ij}(x', x),$$

since f is symmetric, g is anti-symmetric, and the projector π depends quadratically on g .

3. Conservation of H , namely,

$$w(x, x')g(x, x') = 0,$$

identically in (x, x') .

From property 3 it follows that H satisfies the desired Casimir-like condition.

Let us remark that the kernel w could be further generalized preserving the three key properties listed above. One such generalization is

$$w_{ij}(x, x') = f(x, x') \sum_{k,l} \int_{\Omega \times \Omega} \pi_{ik}(x, x_1) G_{kl}(x_1, x_2) \pi_{lj}(x_2, x') dx_1 dx_2,$$

but we shall consider here only the original version of the kernel.

The strong form of the operator is readily obtained via integration by part in the brackets,

$$(A, B) = - \int_{\Omega} A_u(x) \operatorname{div} \left[\int_{\Omega} 2T(x, x') (\nabla B_u(x) - \nabla B_u(x')) dx' \right] dx,$$

and using the expression of T we have

$$(A, B) = - \int_{\Omega} A_u(x) \operatorname{div} \left[\int_{\Omega} M(u(x)) M(u(x')) w(x, x') (\nabla B_u(x) - \nabla B_u(x')) dx' \right] dx.$$

We consider the dynamics for $u(t)$ defined implicitly by

$$\frac{d}{dt} F = (F, S), \quad \text{for all functionals } F(u),$$

where $S(u)$ is an entropy functional of the form

$$S(u) = \int_{\Omega} s(u(x)) dx, \quad s \in C^2(\mathbb{R}).$$

Then we have to choose the function M according to the compatibility condition [Morrison, Physica D, 1986]

$$M(u) \frac{d^2 s}{du^2}(u) = 1.$$

Since $S_u(x) = ds(u(x))/du$, and

$$M(u(x)) \nabla S_u(x) = M(u(x)) \frac{d^2 s}{du^2}(u(x)) \nabla u(x) = \nabla u(x),$$

the symmetric brackets (F, S) become

$$- \int_{\Omega} F_u(x) \operatorname{div} \left[\int_{\Omega} w(x, x') (M(u(x')) \nabla u(x) - M(u(x)) \nabla u(x')) dx' \right] dx.$$

The dynamics of $u(t)$ in strong form then reads,

$$\frac{\partial u}{\partial t} = \operatorname{div} \left[\mathbb{D}(u) \nabla u + \mathbb{F}(u) u \right],$$

which is a Fokker-Planck operator with nonlinear diffusion and friction coefficients given by

$$\mathbb{D}(u) = \int_{\Omega} w(x, x') M(u(x')) dx', \quad \mathbb{F}(u) = - \int_{\Omega} w(x, x') \nabla u(x') dx'.$$

The standard Landau collision operator has this form.

2 Pseudo-spectral implementation. The field u is discretized on a uniform grid of $M \times N$ points on Ω . Hence we work with rectangular arrays $u = u_{ij} \approx u(x_{ij})$ where $x_{ij} = (ih_1, jh_2)$ are the grid points. Arrays of this type are referred to as grid arrays. Vector and tensor arrays have additional indices to accounts for the vector or tensor components.

We make use of a real fft routine, and in the description of the calculations we assume that the fft of a grid array is an array of dimension $M \times \lceil N/2 \rceil$ and such arrays are referred to as dual arrays. The frequencies are denoted by m for the first axis and n for second axis, hence

$$\text{frequencies} = \{(m, n) \mid -\lfloor M/2 \rfloor + 1 \leq m \leq \lfloor M/2 \rfloor, \quad 0 \leq n \leq \lceil N/2 \rceil\}$$

is the set of frequencies, given an $M \times N$ spatial grid.

- Needed functions:

- The function $M(u)$ which corresponds to the desired entropy,
function MFUNC(grid array u)
return grid array $M(u)$
end function
- The function defining the kernel w in terms of (x, x') and g ,
function WFUNC(float x , grid array x' , vector array g)
grid array $f \leftarrow f(x, x')$
tensor array $\pi \leftarrow I - |g|^{-2}(g \otimes g)$
return tensor array $w = f\pi$
end function
- The function defining the derivative of the conserved energy H_u ,
function HUFUNC(grid array u , grid array ϕ , ...)
return grid array $H_u(u, \phi, \dots)$
end function

- Important precomputed arrays and constants:

- De-aliasing filter, a dual array χ ,
Require: int $m_{\text{cut}}, n_{\text{cut}}$, cut-off frequencies.
for $(m, n) \in \text{frequencies}$ **do**
if $|m| \leq m_{\text{cut}}$ and $|n| \leq n_{\text{cut}}$ **then**
 $\chi_{mn} = 1$
else
 $\chi_{mn} = 0.$
end if
end for
- Gradient in Fourier space, a list of dual arrays $\hat{D} = [\hat{D}(1), \hat{D}(2)]$,
Require: Imaginary unit i
for $(m, n) \in \text{frequencies}$ **do**
 $\hat{D}(1)_{mn} \leftarrow im$ $\triangleright x_1$ -derivative
 $\hat{D}(2)_{mn} \leftarrow in$ $\triangleright x_2$ -derivative
end for
- Inverse Laplacian in Fourier space, a dual array $\widehat{\Delta}^{-1}$,

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     $(\widehat{\Delta^{-1}})_{00} \leftarrow 0.$ 
    for  $(m, n) \in \text{frequencies and } (m, n) \neq (0, 0)$  do
         $(\widehat{\Delta^{-1}})_{mn} \leftarrow -(m^2 + n^2)^{-1}.$ 
    end for

    – Normalization of the integrals, float  $\mathcal{N} = \frac{2\pi}{M^2} \frac{2\pi}{N^2}.$ 
    – Multiplicity of frequencies for real fft, dual array  $\mu$ 
        for  $(m, n) \in \text{frequencies}$  do
            if  $n = 0$  then
                 $\mu_{mn} \leftarrow 1.$ 
            else
                 $\mu_{mn} \leftarrow 2.$ 
            end if
        end for

    – Copy of the grid, grid array  $x' = \{x_{ij}\}.$ 

    • Evaluation of the Fokker-Planck operator:
    function COLLISIONOP(grid array  $u$ )
         $\hat{u} \leftarrow \chi \text{fft}(u)$  ▷ Notice de-aliasing with the filter  $\chi$ .
         $\widehat{\nabla u} \leftarrow [\hat{D}(1)\hat{u}, \hat{D}(2)\hat{u}]$ 
         $\phi \leftarrow \text{ifft}(-\widehat{\Delta^{-1}}\hat{u})$ 
        grid array  $h \leftarrow \text{HUFUNC}(u, \phi, \dots)$  ▷ Possibly with arguments.
         $\hat{h} \leftarrow \chi \text{fft}(h)$ 
         $\nabla h \leftarrow \text{ifft}([\hat{D}(1)\hat{h}, \hat{D}(2)\hat{h}])$  ▷ ifft acts only on the frequency axis.
         $M \leftarrow \text{MFUNC}(u)$ 
         $\hat{M} \leftarrow \chi \text{fft}(M)$ 
        for  $x_{ij}$  in spatial grid  $\{x_{ij}\}$  do
            vector array  $g \leftarrow \nabla h(x_{ij}) - \nabla h$ 
            tensor array  $w \leftarrow \text{WFUNC}(x_{ij}, x', g)$ 
             $\hat{w} \leftarrow \text{fft}(w)$  ▷ No need for de-aliasing.
             $\mathbb{D}(x_{ij}) \leftarrow \mathcal{N} \text{sum}(\mu \hat{w} \hat{M})$  ▷ Tensor.
             $\mathbb{F}(x_{ij}) \leftarrow -\mathcal{N} \text{sum}(\mu \hat{w} \widehat{\nabla u})$  ▷ Vector.
        end for
        vector array  $J \leftarrow \mathbb{D} \text{ifft}(\widehat{\nabla u}) + \mathbb{F}M$ 
         $\hat{J} = \text{fft}([J(1), J(2)])$  ▷ Component-wise fft, no de-aliasing.
         $\text{div } J \leftarrow \text{ifft}(\hat{D} \cdot \hat{J})$  ▷ Vector-vector multiplication.
        return grid array  $\text{div } J$ 
    end function

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